

# LeadSpark CRM

Complete Architecture Diagram & System Reference

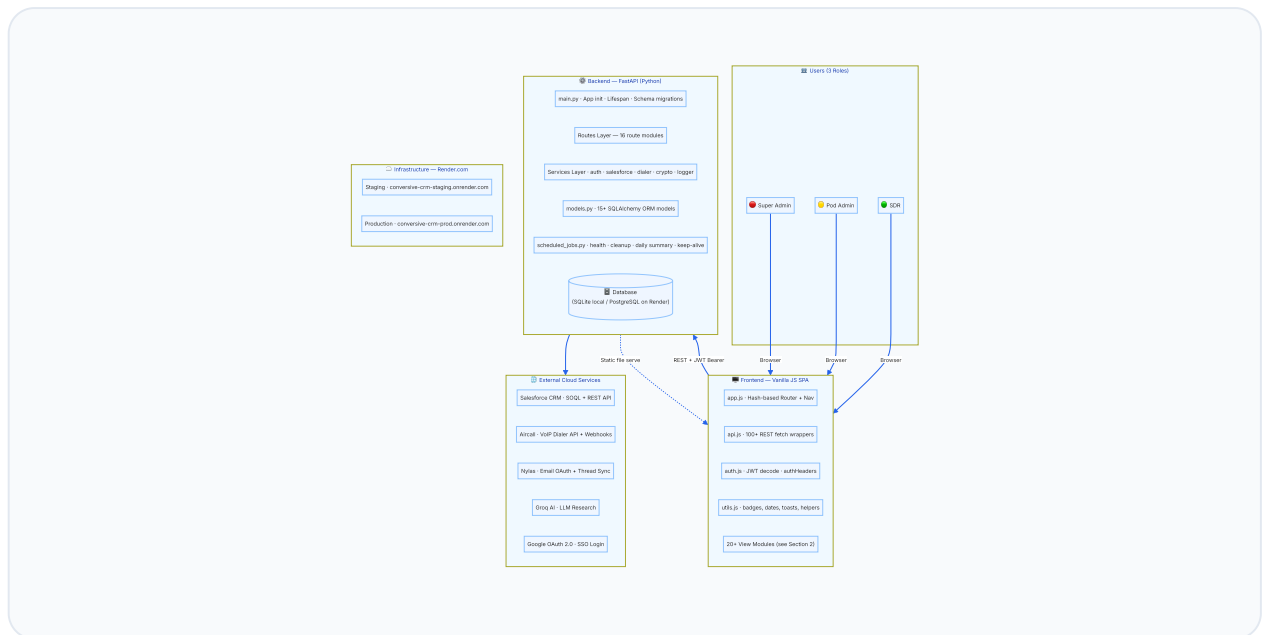
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|----------------------------------------------------|------------------------------------|-----------------------------------------------|-----------------------------------------------|
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# 1 System Overview

LeadSpark CRM is a full-stack application. A Vanilla JS Single-Page Application (SPA) communicates with a FastAPI (Python) backend over REST/JWT. The backend connects to a relational database and five external cloud services. Both environments are deployed on Render.com and triggered automatically by Git branch changes.



The frontend is a hash-router SPA with no build step — pure HTML, CSS, and ES modules. All API calls are centralized in `api.js`. Authentication state is stored in `localStorage` as a JWT token. Views are loaded dynamically on navigation.

### Core Module Layer

| File                  | Responsibility                                                                                                                               | Key Exports                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>app.js</code>   | Hash-based router, nav active state, top-level call action handler                                                                           | <code>loadView</code> , <code>_markActive</code> , <code>handleCallAction</code> , <code>_parseHash</code>                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>api.js</code>   | All network calls — wraps fetch with auth headers + error handling. 100+ endpoints.                                                          | <code>fetchLeads</code> , <code>fetchLead</code> , <code>patchLead</code> , <code>updateLeadStatus</code> , <code>closeLead</code> , <code>markNoShow</code> , <code>createLead</code> , <code>patchLeadResearch</code> , <code>logCall</code> , <code>fetchAdminUsers</code> , <code>bulkAssign</code> , <code>autoAssignAll</code> , <code>syncSalesforce</code> , <code>fetchPods</code> , <code>fetchSfLogs</code> , <code>sendEmail</code> , <code>startDialerCall</code> , <code>reprioritizeLead</code> , <code>fetchPendingTasks</code> , ... |
| <code>auth.js</code>  | JWT decode, role derivation ( <code>isSuperAdmin</code> / <code>isPodAdmin</code> / <code>isSDR</code> ), auth headers, URL token extraction | <code>parseJwt</code> , <code>authHeaders</code> , <code>deriveRoles</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>utils.js</code> | Shared UI helpers used by all views                                                                                                          | <code>statusBadgeClass</code> , <code>displayStatus</code> , <code>fmtDate</code> , <code>fmtDateTime</code> , <code>fmtRelativeTime</code> , <code>timeInStatus</code> , <code>fullName</code> , <code>showToast</code> , <code>showLoader</code> , <code>renderPaginated</code> , <code>ssoTag</code>                                                                                                                                                                                                                                               |

### View Modules — </frontend/js/views/>

| View File                 | Screen / Feature                                  | Key Functions                                                                                                                                  |
|---------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>dashboard.js</code> | Dashboard — status KPI cards + recent leads table | <code>renderDashboard</code>                                                                                                                   |
| <code>lead_list.js</code> | All Leads / My Leads table with filters,          | <code>renderLeads</code> , <code>loadPage</code> , <code>_renderRows</code> , <code>_renderPagination</code> , <code>_getPriorityTier</code> , |

| View File                          | Screen / Feature                                                                           | Key Functions                                                                                                                                              |
|------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    | search, pagination, outcome filter, priority badges                                        | _priorityBadge                                                                                                                                             |
| <code>lead_detail.js</code>        | Lead detail page — profile card, research panel, notes, tasks, attachments, status stepper | renderLeadDetail, renderCallLogs, renderNotes, renderTasks, updateResearchProgress, countResearchFilled, setOutcome, bindTaskEvents, _renderEnrichmentCard |
| <code>lead_kanban.js</code>        | Kanban board with drag-and-drop status columns                                             | renderKanban, _bindDragAndDrop, _updateCounts, _getDragAfterElement                                                                                        |
| <code>lead_emails_tab.js</code>    | Email thread viewer and composer (Nylas-powered)                                           | loadEmailsTab, _groupIntoThreads, _renderSplitPane, _renderThreadView, _renderComposeView, _bindReplyEvents, _showCompose                                  |
| <code>lead_calls_tab.js</code>     | Calls history tab per lead (manual + Aircall)                                              | loadCallsTab, _fmtDur, _fmtLastCalled, _outcomeBadge                                                                                                       |
| <code>lead_messaging_tab.js</code> | Embedded messaging iframe widget                                                           | loadMessagingTab                                                                                                                                           |
| <code>lead_helpers.js</code>       | Shared lead UI utilities                                                                   | canModifyLead, renderSourceBadge, formatNoteDate                                                                                                           |
| <code>admin.js</code>              | Admin Panel — user list with roles, last login, dialer flag toggles                        | renderAdmin, renderBoard, _renderRow, _activityBadge                                                                                                       |
| <code>assignments.js</code>        | Lead assignment table — unassigned queue, bulk assign/delete                               | renderAssignments, renderAssignedTable, updateActionBar, _doBulkDelete                                                                                     |
| <code>Pods.js</code>               | Pod management — create/delete pods,                                                       | renderPods                                                                                                                                                 |

| View File                           | Screen / Feature                                                        | Key Functions                                                                                                                                                                            |
|-------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | add/remove SDR members                                                  |                                                                                                                                                                                          |
| <code>settings.js</code>            | Settings hub — tabbed: Connection, Sync, AI Dialer                      | <code>renderSettings</code>                                                                                                                                                              |
| <code>settings_connection.js</code> | Salesforce connection status + credential form                          | <code>connectionTabHTML</code> , <code>bindConnectionTab</code>                                                                                                                          |
| <code>settings_sync.js</code>       | Sync direction, lead limit, multi-pod SDR flag                          | <code>bindSyncTab</code>                                                                                                                                                                 |
| <code>settings_ai_dialer.js</code>  | Aircall API credentials config                                          | <code>bindAiDialerTab</code> , <code>_toggleDialerCreds</code>                                                                                                                           |
| <code>leaderboard.js</code>         | SDR leaderboard with time period presets                                | <code>renderLeaderboard</code>                                                                                                                                                           |
| <code>metrics.js</code>             | SDR metrics — KPI cards, daily trend chart, per-SDR table               | <code>renderMetrics</code> , <code>loadKpiCards</code> , <code>loadTrendChart</code> , <code>loadSdrTable</code>                                                                         |
| <code>sdr_performance.js</code>     | Individual SDR performance view                                         | <code>renderSdrPerformance</code> , <code>_metricCard</code>                                                                                                                             |
| <code>activity_feed.js</code>       | Org-wide activity feed — calls, emails, status changes, research events | <code>renderActivityFeed</code> , <code>_renderRow</code> , <code>_renderDetailPanel</code> , <code>_renderCallDetail</code> , <code>_renderEmailDetail</code> , <code>_exportCsv</code> |
| <code>audit_logs.js</code>          | Tabbed audit log — Activity, Logins, Feedback, SF Logs                  | <code>renderAuditLogs</code> , <code>loadTab</code> , <code>loadActivity</code> , <code>loadLogins</code> , <code>loadFeedback</code> , <code>loadSfLogs</code>                          |
| <code>sf_logs.js</code>             | Salesforce operation log list + detail drawer                           | <code>renderSfLogs</code> , <code>applyFilters</code> , <code>_loadLogs</code> , <code>_renderLogRow</code> , <code>_showLogDetail</code>                                                |
| <code>task_notifications.js</code>  | Overdue task notification bell —                                        | <code>initTaskNotifications</code> , <code>destroyTaskNotifications</code> , <code>_pollTasks</code> ,                                                                                   |

| View File                     | Screen / Feature                                                                | Key Functions                                                                                                                      |
|-------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
|                               | poll, sound chime,<br>push notification                                         | _updateBadge, _renderPanel,<br>_sendPushNotification                                                                               |
| <code>modals.js</code>        | All modal dialogs —<br>Call log, New Lead,<br>Add User, Bulk<br>Import, No-Show | initModals, openCallModal,<br>_initCallModal, _initNewLeadModal,<br>_initAddUserModal,<br>_initBulkImportModal,<br>openNoShowModal |
| <code>upload.js</code>        | Upload Center —<br>tabs for CSV and<br>Google Sheets,<br>upload history         | renderUpload, _renderHistory,<br>_showStep, _renderResultBanner                                                                    |
| <code>upload_csv.js</code>    | CSV upload flow —<br>file pick, column<br>mapping, preview,<br>import           | bindCsvEvents, handleFile, showFileCard                                                                                            |
| <code>upload_gsheet.js</code> | Google Sheets import<br>flow — URL input,<br>preview, import                    | bindGSheetEvents                                                                                                                   |
| <code>my_calls.js</code>      | SDR's own call<br>history grouped by<br>date                                    | renderMyCalls, renderStats, renderBody,<br>_groupCalls, _callCard                                                                  |
| <code>user_guide.js</code>    | In-app User Guide +<br>What's New release<br>notes                              | renderUserGuide                                                                                                                    |
| <code>sdr_settings.js</code>  | SDR personal settings                                                           | renderSdrSettings                                                                                                                  |

FastAPI application with a clean layered architecture: Entry → Routes → Services → Models. Schema migrations run automatically on startup via `migrations.py`. Background jobs are started during the app lifespan.

### Entry Point — `main.py`

| Function                                    | Purpose                                                                        |
|---------------------------------------------|--------------------------------------------------------------------------------|
| <code>lifespan()</code>                     | Startup: run migrations, start scheduled jobs. Shutdown: stop jobs gracefully. |
| <code>_get_or_create_sync_settings()</code> | Ensure default SyncSettings row exists on first boot                           |
| <code>_repair_truncated_emails()</code>     | Background repair pass for email bodies truncated in earlier versions          |
| <code>health_check()</code>                 | GET /health — returns system status                                            |
| <code>read_root()</code>                    | GET / — serves frontend index.html                                             |

### Routes Layer — `/backend/routes/`

| Route File                  | Prefix     | Key Endpoints                                                                                                                                                                                                                                                                             |
|-----------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>auth_routes.py</code> | /auth      | login (Google OAuth URL), auth_callback (exchange code → JWT), me (current user), logout, heartbeat, get_config, demo_login                                                                                                                                                               |
| <code>lead_routes.py</code> | /api/leads | GET list (paginated + filters), GET /:id, POST (create), PATCH /:id, POST /kanban/move, GET /kanban, GET /dashboard-stats, GET /activity-feed, POST /:id/calls, POST /:id/close, POST /:id/no-show, PATCH /:id/outcome, PATCH /:id/research, PATCH /:id/priority, GET /:id/status-history |

| Route File                         | Prefix                     | Key Endpoints                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>admin_routes.py</code>       | <code>/api/admin</code>    | GET/POST <code>/users</code> , DELETE <code>/users/:id</code> , PATCH <code>/users/:id/role</code> , PATCH <code>/users/:id/access</code> , PATCH <code>/users/:id/settings</code> , GET/POST <code>/unassigned</code> , POST <code>/bulk-assign</code> , POST <code>/auto-assign</code> , POST <code>/unassign</code> , GET/PATCH <code>/sync-settings</code> , POST <code>/upload-preview</code> , POST <code>/upload-sheet</code> , POST <code>/gsheet-preview</code> , POST <code>/gsheet-import</code> , GET <code>/upload-logs</code> , POST <code>/upload-sdr-csv</code> |
| <code>pod_routes.py</code>         | <code>/api/pods</code>     | GET <code>/</code> , POST <code>/</code> , PATCH <code>/:id</code> , DELETE <code>/:id</code> , POST <code>/:id/members</code> , DELETE <code>/:id/members/:uid</code> , POST <code>/:id/assign-leads</code>                                                                                                                                                                                                                                                                                                                                                                    |
| <code>email_routes.py</code>       | <code>/api/email</code>    | GET/POST <code>/nylas-config</code> , GET <code>/auth-url</code> , GET <code>/callback</code> , GET <code>/status</code> , DELETE <code>/disconnect</code> , POST <code>/send</code> , GET <code>/leads/:id/emails</code> , DELETE <code>/repair</code> , GET <code>/download/:id</code>                                                                                                                                                                                                                                                                                        |
| <code>dialer_routes.py</code>      | <code>/api/dialer</code>   | POST <code>/start-call</code> , POST <code>/webhook</code> , GET <code>/status</code> , GET/PATCH <code>/config</code> , POST <code>/test-connection</code> , GET <code>/users</code> , GET <code>/numbers</code> , PATCH <code>/calls/:id/outcome</code> , GET <code>/debug/calls</code> , GET <code>/debug/calls/:id</code> , POST <code>/debug/repair-orphaned</code>                                                                                                                                                                                                        |
| <code>ai_research_routes.py</code> | <code>/api/research</code> | POST <code>/leads/:id</code> — triggers Groq LLM, sanitizes response, saves research fields to lead                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

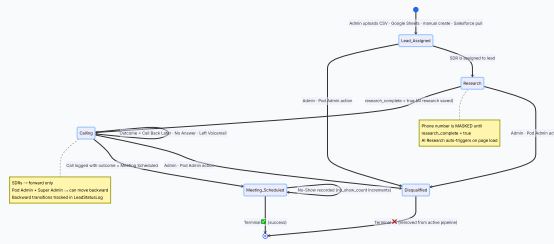
| Route File                             | Prefix                                  | Key Endpoints                                                                                                                                                                                                       |
|----------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>sf_connection_routes.py</code>   | <code>/api/salesforce</code>            | GET <code>/status</code> , POST <code>/connect</code> , DELETE <code>/disconnect</code> , POST <code>/reconnect</code> , GET <code>/record-types</code> , POST <code>/sync</code>                                   |
| <code>sf_log_routes.py</code>          | <code>/api/sf-logs</code>               | GET <code>/</code> (paginated + filters), GET <code>/:id</code> , GET <code>/export-csv</code>                                                                                                                      |
| <code>note_routes.py</code>            | <code>/api/leads/:id/notes</code>       | GET <code>/</code> , POST <code>/</code> , DELETE <code>/:notelid</code>                                                                                                                                            |
| <code>task_routes.py</code>            | <code>/api/leads/:id/tasks</code>       | GET <code>/</code> , POST <code>/</code> , PATCH <code>/:taskId/toggle</code> , DELETE <code>/:taskId</code> , GET <code>/pending</code> , PATCH <code>/:taskId/snooze</code> , PATCH <code>/:taskId/dismiss</code> |
| <code>attachment_routes.py</code>      | <code>/api/leads/:id/attachments</code> | GET <code>/</code> , POST <code>/upload</code> , GET <code>/:attid/download</code> , DELETE <code>/:attid</code>                                                                                                    |
| <code>leaderboard_routes.py</code>     | <code>/api/leaderboard</code>           | GET <code>/</code> — SDR rankings by meetings, calls, disqualified                                                                                                                                                  |
| <code>metrics_routes.py</code>         | <code>/api/metrics</code>               | GET <code>/summary</code> , GET <code>/daily-trend</code> , GET <code>/sdr-table</code> , GET <code>/export-csv</code> , POST <code>/seed</code> (dev only)                                                         |
| <code>sdr_performance_routes.py</code> | <code>/api/sdr</code>                   | GET <code>/performance/:id</code> , GET <code>/call-summary</code>                                                                                                                                                  |
| <code>search_routes.py</code>          | <code>/api/search</code>                | GET <code>/</code> — global search across leads + users (role-aware)                                                                                                                                                |
| <code>webhook_routes.py</code>         | <code>/api/webhooks</code>              | GET <code>/nylas</code> (challenge verification), POST <code>/nylas</code> (inbound email events → thread sync)                                                                                                     |

## Services / Core Logic Layer

| File                             | Responsibility                                                                    | Key Functions                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>auth.py</code>             | JWT creation/decoding, Google OAuth handshake, role guard decorators              | <code>create_jwt</code> , <code>decode_jwt</code> , <code>get_current_user</code> , <code>require_admin</code> , <code>require_super_admin</code> , <code>require_pod_admin_or_above</code> , <code>google_auth_url</code> , <code>exchange_code_for_user</code>                                                                                                                                                   |
| <code>salesforce.py</code>       | Full Salesforce integration — pull/push leads, sync SDRs, build lead descriptions | <code>get_sf_client</code> , <code>sync_leads_from_salesforce</code> , <code>push_pending_leads_to_salesforce</code> , <code>push_lead_to_salesforce</code> , <code>create_new_lead_in_salesforce</code> , <code>sync_sdrs_from_salesforce</code> , <code>push_sdr_metrics_to_salesforce</code> , <code>_build_lead_description</code> , <code>_build_query</code> , <code>get_record_types_from_salesforce</code> |
| <code>dialer_service.py</code>   | Provider-agnostic dialer service — routes to concrete provider                    | <code>get_active_provider</code> , <code>initiate_call</code> , <code>handle_webhook</code> , <code>test_provider_connection</code> , <code>get_provider_users</code> , <code>get_provider_numbers</code> , <code>get_dialer_config</code> , <code>save_dialer_config</code>                                                                                                                                       |
| <code>dialer_provider.py</code>  | Abstract base class defining the dialer provider interface                        | <code>DialerProvider</code> (ABC), <code>CallEventType</code> (enum), <code>NormalizedCallEvent</code> , <code>InitiateCallResult</code>                                                                                                                                                                                                                                                                           |
| <code>aircall_provider.py</code> | Concrete Aircall implementation — REST calls + webhook parsing                    | <code>AircallDialerProvider</code> , <code>initiate_call</code> , <code>handle_webhook</code> , <code>fetch_call</code> , <code>fetch_transcript</code> , <code>get_users</code> , <code>get_numbers</code> , <code>test_connection</code> , <code>_format_e164</code> , <code>_find_user_by_email</code>                                                                                                          |
| <code>sf_logger.py</code>        | Log every Salesforce operation to DB with success/failure status                  | <code>log_sf_operation</code> , <code>_safe_json</code>                                                                                                                                                                                                                                                                                                                                                            |
| <code>activity_logger.py</code>  | Log SDR activity events (calls, emails, status changes, research)                 | <code>log_activity</code>                                                                                                                                                                                                                                                                                                                                                                                          |

| File                             | Responsibility                                               | Key Functions                                                                                                                                                                                                    |
|----------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>crypto.py</code>           | Fernet symmetric encryption for storing OAuth tokens at rest | <code>encrypt_token</code> , <code>decrypt_token</code> , <code>_get_key</code>                                                                                                                                  |
| <code>access_db.py</code>        | Allowlist of permitted user emails (CSV manageable)          | <code>is_user_allowed</code> , <code>add_allowed_user</code> , <code>remove_allowed_user</code> , <code>list_allowed_users</code> , <code>process_csv</code>                                                     |
| <code>audience_manager.py</code> | Sync leads to external contact/audience manager              | <code>search_contact</code> , <code>create_contact</code> , <code>ensure_contact</code> , <code>sync_leads_to_am_background</code>                                                                               |
| <code>scheduled_jobs.py</code>   | Background recurring jobs managed via APScheduler            | <code>start_scheduled_jobs</code> , <code>stop_scheduled_jobs</code> , <code>_health_check_job</code> , <code>_aggregate_daily_summaries</code> , <code>_cleanup_old_logs</code> , <code>_keep_alive_ping</code> |
| <code>migrations.py</code>       | Idempotent schema migrations run on each startup             | <code>run_schema_migrations</code> , <code>_column_exists</code> , <code>_table_exists</code>                                                                                                                    |
| <code>database.py</code>         | SQLAlchemy engine + session factory                          | <code>get_db</code>                                                                                                                                                                                              |

Every lead progresses through a strict pipeline. Certain transitions are gated on preconditions (research complete, call logged). Terminal states cannot be exited.



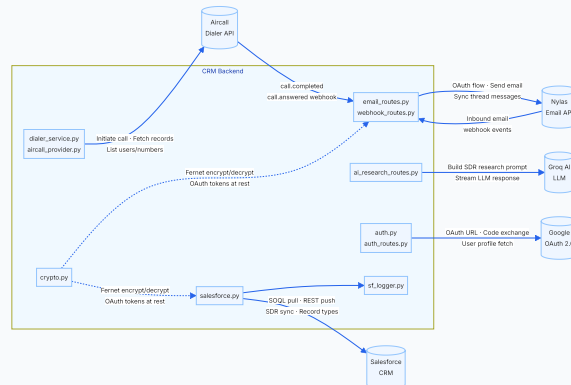
### Status Values (Status enum)

| Status            | Type         | Description                                               |
|-------------------|--------------|-----------------------------------------------------------|
| Lead Assigned     | Active       | Lead exists in pool, awaiting SDR action                  |
| Research          | Active       | SDR has been assigned; AI research auto-triggered         |
| Calling           | Active       | Research complete; SDR actively calling                   |
| Meeting Scheduled | Terminal ✓   | Call resulted in a meeting; tracked in leaderboard        |
| Disqualified      | Terminal ✗   | Lead removed from pipeline by admin action                |
| Lead Unassigned   | Display only | Frontend display alias for unassigned Lead Assigned leads |

### Call Outcome Values (CallOutcome enum)

| Outcome           | Effect on Lead                                               |
|-------------------|--------------------------------------------------------------|
| Meeting Scheduled | Triggers auto-transition to Meeting Scheduled status         |
| Meeting Confirmed | Escalation of Meeting Scheduled — company marked as resolved |
| Call Back Later   | Lead stays in Calling; priority auto-decremented             |

| Outcome        | Effect on Lead                                       |
|----------------|------------------------------------------------------|
| No Answer      | Lead stays in Calling; auto-deprioritization applied |
| Left Voicemail | Lead stays in Calling                                |
| Not Interested | Lead stays in Calling (admin can disqualify)         |
| Wrong Number   | Lead stays in Calling (admin can disqualify)         |



| Service           | Auth Mechanism                                          | Data Flow                                                                                                         | Key Backend Files                                                                                       |
|-------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Salesforce</b> | Username + Password + Security Token (stored encrypted) | Pull leads on sync · Push status updates + SDR metrics · Two-way: CRM → Salesforce, Salesforce → CRM              | <code>salesforce.py</code> ,<br><code>sf_connection_routes.py</code> ,<br><code>sf_logger.py</code>     |
| <b>Aircall</b>    | API Key + API ID (stored encrypted)                     | Initiate outbound calls · Receive webhook events (call.completed, call.answered) · Fetch recordings + transcripts | <code>aircall_provider.py</code> ,<br><code>dialer_service.py</code> ,<br><code>dialer_routes.py</code> |
| <b>Nylas</b>      | OAuth 2.0 per-user grant + API Key (encrypted)          | Connect SDR email → Send emails → Sync inbound thread replies → Inbound webhook for real-time delivery            | <code>email_routes.py</code> ,<br><code>webhook_routes.py</code> ,<br><code>crypto.py</code>            |
| <b>Groq AI</b>    | API Key (env variable)                                  | Build research prompt from lead profile → Stream LLM response → Sanitize + save structured research fields        | <code>ai_research_routes.py</code>                                                                      |

| Service      | Auth Mechanism               | Data Flow                                                                     | Key Backend Files                                  |
|--------------|------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------|
| Google OAuth | OAuth 2.0 client credentials | Redirect to Google consent → Exchange code → Get user profile → Issue CRM JWT | <code>auth.py</code> , <code>auth_routes.py</code> |

Role guards are enforced server-side on every endpoint using FastAPI dependency injection:

`require_super_admin`, `require_admin` (super + pod), `require_pod_admin_or_above`.

| Capability                                      | Super Admin | Pod Admin   | SDR |
|-------------------------------------------------|-------------|-------------|-----|
| Upload leads (CSV / Google Sheets)              | ✓           | ✗           | ✗   |
| Manage all users (create, delete, role, access) | ✓           | ✗           | ✗   |
| Toggle dialer / email-sync flags per user       | ✓           | ✗           | ✗   |
| Configure Salesforce connection + credentials   | ✓           | ✗           | ✗   |
| View SF Audit Logs + export CSV                 | ✓           | ✗           | ✗   |
| Delete leads permanently                        | ✓           | ✗           | ✗   |
| Create / delete pods                            | ✓           | ✗           | ✗   |
| View all pods + all leads (org-wide)            | ✓           | Pod only    | ✗   |
| Assign / unassign leads in pod                  | ✓           | ✓           | ✗   |
| Move leads backward in Kanban                   | ✓           | ✓           | ✗   |
| Disqualify leads                                | ✓           | ✓           | ✗   |
| SDR performance/metrics dashboard               | ✓           | ✓           | ✗   |
| Manage pod members (add/remove SDRs)            | ✓           | ✓ (own pod) | ✗   |
| View my assigned leads                          | ✓           | ✓           | ✓   |
| AI Research auto-trigger + save                 | ✓           | ✓           | ✓   |
| Log calls + set call outcomes                   | ✓           | ✓           | ✓   |
| Add / delete notes on leads                     | ✓           | ✓           | ✓   |

| Capability                                 | Super Admin | Pod Admin | SDR            |
|--------------------------------------------|-------------|-----------|----------------|
| Create / complete / snooze / dismiss tasks | ✓           | ✓         | ✓              |
| Email leads via Nylas (compose + reply)    | ✓           | ✓         | ✓              |
| Upload/download attachments on leads       | ✓           | ✓         | ✓              |
| Initiate Aircall dialer calls              | ✓           | ✓         | ✓ (if enabled) |
| View leaderboard                           | ✓           | ✓         | ✓              |
| View my calls + call history               | ✓           | ✓         | ✓              |
| Access User Guide + release notes          | ✓           | ✓         | ✓              |

All models are SQLAlchemy ORM classes defined in `backend/models.py`. Schema changes are applied via incremental migrations in `migrations.py`.

| Model         | Key Fields                                                                                                                                                                                         | Relationships                                                                                 |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| User          | id (UUID), email, name, role (super_admin/pod_admin/sdr), is_active, last_login, dialer_enabled, email_sync_enabled, google_id                                                                     | leads (assigned), pod_memberships, mailbox, activity_logs                                     |
| Lead          | id (UUID), first_name, last_name, email, phone, company, title, linkedin_url, source, status, priority_score, assigned_to_id, research_complete, no_show_count, created_at, updated_at, sf_lead_id | assigned_user, notes, tasks, call_logs, status_logs, attachments, email_threads, dialer_calls |
| Pod           | id (UUID), name, admin_id, created_at                                                                                                                                                              | members (Users), admin (User)                                                                 |
| CallLog       | id (UUID), lead_id, user_id, outcome, notes, duration_seconds, called_at, source                                                                                                                   | lead, user                                                                                    |
| DialerCall    | id, lead_id, user_id, provider, provider_call_id, status, outcome, duration, recording_url, transcript_url, started_at, ended_at                                                                   | lead, user                                                                                    |
| Note          | id (UUID), lead_id, content, author, created_at                                                                                                                                                    | lead                                                                                          |
| Task          | id (UUID), lead_id, user_id, title, due_date, due_time, done, dismissed, snoozed_until, created_at                                                                                                 | lead, user                                                                                    |
| LeadStatusLog | id, lead_id, from_status, to_status, changed_by, changed_at                                                                                                                                        | lead                                                                                          |

| Model                           | Key Fields                                                                                                                        | Relationships  |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>SyncSettings</b>             | id, direction (push/pull/both),<br>lead_limit, record_type_id,<br>allow_multi_pod_sdr, updated_at                                 | —              |
| <b>SalesforceIntegrationLog</b> | id, operation_type, status<br>(success/failure), lead_id, sf_id,<br>payload_sent, payload_received,<br>error_message, created_at  | —              |
| <b>SalesforceConnection</b>     | id, username, instance_url,<br>token_encrypted, connected_at                                                                      | —              |
| <b>NylasConfig</b>              | id, client_id, api_key_encrypted,<br>redirect_uri, created_at                                                                     | —              |
| <b>UserMailbox</b>              | id, user_id, email, grant_id,<br>access_token_encrypted,<br>connected_at                                                          | user           |
| <b>LeadEmailActivity</b>        | id, lead_id, thread_id, message_id,<br>direction (inbound/outbound), subject,<br>body, from_email, to_email, sent_at,<br>provider | lead, thread   |
| <b>EmailThread</b>              | id, lead_id, provider_thread_id,<br>subject, last_message_at                                                                      | lead, messages |
| <b>LeadAttachment</b>           | id, lead_id, filename,<br>original_filename, file_path, file_size,<br>mime_type, uploaded_by, uploaded_at                         | lead           |
| <b>LeadUploadLog</b>            | id, filename, rows_total,<br>rows_imported, rows_skipped,<br>rows_updated, source_label,<br>uploaded_by, created_at               | —              |
| <b>LoginLog</b>                 | id, user_id, email, ip_address,<br>user_agent, logged_in_at                                                                       | —              |
| <b>UserActivityLog</b>          | id, user_id, activity_type, lead_id,<br>metadata_json, created_at                                                                 | —              |

| Model                    | Key Fields                                                                  | Relationships |
|--------------------------|-----------------------------------------------------------------------------|---------------|
| UserActivityDailySummary | id, user_id, date, calls_made, emails_sent, leads_moved, meetings_scheduled | —             |
| AllowedUser              | id, email (case-insensitive), added_at                                      | —             |
| Feedback                 | id, user_id, content, status, created_at                                    | —             |

The codebase is partitioned into 8 communities detected by the **code-review-graph** knowledge graph using the Leiden community detection algorithm. There are **312 cross-community edges**.

| Community            | Nodes | Lang       | Cohesion | Description                                                                                                                                                                                                                                                |
|----------------------|-------|------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tests-<br>Lead       | 556   | Python     | 0.36     | Full backend Python test suite. Covers: access_db, admin routes, attachment routes, auth, company resolution, email routes, lead routes, leaderboard, models, notes, outcome filters, pod routes, priority, salesforce, search, SF logs, SF logger, tasks. |
| views-<br>render     | 326   | JavaScript | 0.11     | All 20+ frontend JS view modules in /frontend/js/views/. Each view module is responsible for rendering one screen of the SPA.                                                                                                                              |
| routes-<br>Lead      | 182   | Python     | 0.04     | All 16 FastAPI route handler modules in /backend/routes/. These are the API surface of the application.                                                                                                                                                    |
| tests-<br>it:returns | 146   | JavaScript | 0.17     | Frontend JS unit tests using Jest: api.test.js (80+ API wrapper tests), auth.test.js (JWT + role derivation), utils.test.js (badge, date format, toast tests).                                                                                             |
| js-fetch             | 127   | JavaScript | 0.14     | Core JS modules: api.js (100+ fetch wrappers), auth.js (JWT decode), utils.js (18+ shared helpers), app.js (hash router logic).                                                                                                                            |
| backend-<br>user     | 123   | Python     | 0.08     | Core Python backend logic outside of routes: models.py, auth.py, salesforce.py, aircall_provider.py, dialer_service.py, scheduled_jobs.py, migrations.py, crypto.py, sf_logger.py, activity_logger.py, access_db.py, main.py.                              |

| Community         | Nodes | Lang       | Cohesion | Description                                                                                                                                                                                               |
|-------------------|-------|------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tests-<br>shows   | 102   | JavaScript | 0.07     | End-to-end Playwright test suites: sdr-flow.spec.js (14 tests), pod-admin-flow.spec.js (16 tests), super-admin-flow.spec.js (18 tests), task-and-outcome-filter.spec.js (20+ tests against live staging). |
| helpers-<br>token | 4     | JavaScript | 0.04     | Shared test helpers in /tests/helpers/. Small utility functions for E2E test scaffolding.                                                                                                                 |

Key Cross-Community Relationships

| From Community   | To Community | Edge Type     | Example                                                                                            |
|------------------|--------------|---------------|----------------------------------------------------------------------------------------------------|
| routes-lead      | backend-user | CALLS         | auth_routes.py → auth.py::google_auth_url;<br>email_routes.py → crypto.py::encrypt_token           |
| tests-lead       | backend-user | CALLS (tests) | test_auth.py → auth.py::create_jwt;<br>test_salesforce.py → salesforce.py::_build_lead_description |
| tests-lead       | routes-lead  | CALLS (tests) | test_lead_routes.py → lead_routes endpoints                                                        |
| tests-it:returns | js-fetch     | CALLS (tests) | api.test.js → api.js::fetchLeads; auth.test.js → auth.js::parseJwt                                 |
| tests-shows      | views-render | E2E (browser) | sdr-flow.spec.js exercises lead_list.js, lead_detail.js, kanban views                              |

## Deployment Pipeline

⚠ **Every push to develop or main triggers a Render build and consumes paid pipeline minutes.**

Always batch related changes into a single commit. Never auto-push — always ask the user for approval before each push.

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## Pre-Production Checklist

| # | Check                                                                        | Required    |
|---|------------------------------------------------------------------------------|-------------|
| 1 | Feature deployed and tested on Staging                                       | ✓ Mandatory |
| 2 | No critical bugs on Staging                                                  | ✓ Mandatory |
| 3 | Logs are working (SF logs, activity logs, login logs)                        | ✓ Mandatory |
| 4 | <code>detect_changes</code> reviewed — no unaddressed HIGH-risk items        | ✓ Mandatory |
| 5 | <code>get_impact_radius</code> run — blast radius is acceptable              | ✓ Mandatory |
| 6 | <code>get_affected_flows</code> run — impacted flows are tested              | ✓ Mandatory |
| 7 | User Guide updated with screenshots in <code>docs/user_guide.md</code>       | ✓ Mandatory |
| 8 | Release notes updated in <code>docs/release-notes.md</code>                  | ✓ Mandatory |
| 9 | In-app release notes updated in <code>frontend/js/views/user_guide.js</code> | ✓ Mandatory |

## Branch → Environment Mapping

| Git Branch           | Environment | URL                                 | Purpose                                          |
|----------------------|-------------|-------------------------------------|--------------------------------------------------|
| deve <sup>l</sup> op | Staging     | conversive-crm-staging.onrender.com | All development, testing, debugging, experiments |

| Git Branch        | Environment | URL                              | Purpose                                              |
|-------------------|-------------|----------------------------------|------------------------------------------------------|
| <code>main</code> | Production  | conversive-crm-prod.onrender.com | Production — explicit approval only, never auto-push |

Per `AGENT.md` — the following systems are core to business functionality. They MUST NOT be modified without running `get_impact_radius` first. Any change must include logging, be tested on staging, and preserve all existing behavior.

 **Rule:** If in doubt about modifying any of these systems → STOP and ask for clarification before proceeding.

| System                             | Key Files                                                                                                                        | Risk                                                                                              | Rules                                                                                                                                      |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Salesforce Sync</b>             | <code>salesforce.py</code><br><code>sf_logger.py</code><br><code>sf_connection_routes.py</code><br><code>sf_log_routes.py</code> |  <b>HIGH</b>     | All SF operations must log success/failure via <code>sf_logger</code> . Test push/pull on staging with real credentials before production. |
| <b>Lead Assignment Logic</b>       | <code>admin_routes.py</code><br><code>pod_routes.py</code>                                                                       |  <b>HIGH</b>     | Round-robin distribution, <code>active_lead_cap</code> , pod integrity constraints must all be preserved.                                  |
| <b>Authentication (Google SSO)</b> | <code>auth.py</code><br><code>auth_routes.py</code><br><code>access_db.py</code>                                                 |  <b>HIGH</b>   | JWT secrets in env vars. Role guards enforced server-side. Allowlist checked on login. Never expose decode logic client-side.              |
| <b>Logs Dashboard</b>              | <code>sf_log_routes.py</code><br><code>audit_logs.js</code><br><code>activity_logger.py</code>                                   |  <b>MEDIUM</b> | All operations must remain traceable. Do not delete logs without admin approval.                                                           |
| <b>Dialer Integration</b>          | <code>dialer_service.py</code><br><code>aircall_provider.py</code><br><code>dialer_routes.py</code>                              |  <b>MEDIUM</b> | Provider abstraction must remain intact. Webhook signature verification must not be removed.                                               |
| <b>Email Integration (Nylas)</b>   | <code>email_routes.py</code><br><code>webhook_routes.py</code><br><code>crypto.py</code>                                         |  <b>MEDIUM</b> | OAuth tokens encrypted at rest via Fernet. Never log raw access tokens. Webhook challenge                                                  |

| System               | Key Files             | Risk                             | Rules                                                                                                             |
|----------------------|-----------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------|
|                      |                       |                                  | response must remain intact.                                                                                      |
| AI Research Pipeline | ai_research_routes.py | <div><div></div>LOW-MEDIUM</div> | research_complete flag gates phone masking. Sanitize LLM output before saving. Rate-limit awareness for Groq API. |

Scheduled Jobs — Do Not Remove

| Job                        | Impact if Removed                                                  |
|----------------------------|--------------------------------------------------------------------|
| _health_check_job          | Loss of system health monitoring; silent failures                  |
| _aggregate_daily_summaries | SDR metrics dashboard data stops updating                          |
| _cleanup_old_logs          | Database grows unbounded; performance degradation                  |
| _keep_alive_ping           | Render service cold-starts; 30s+ response delays for first request |